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## Geochemical Correlation of Gas and Condensate from The Lower Seurula Reservoir with Potential Source Rocks in The North Sumatra Basin

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**ABSTRACT** - Recent exploration activities in Block “B” of the North Sumatra Basin, including the acquisition and processing of 610 km<sup>2</sup> 3D seismic data followed by a two-well exploratory drilling campaign, lead to gas and condensate discoveries within the Late Miocene sandstone reservoirs of the Lower Seurula Formation. The campaign targets two structures: Arun Overburden (AOB), located above the Arun Structure, and Rayeu. Previous studies focus primarily on the giant Arun and SLS Fields due to their substantial hydrocarbon reserves. Consequently, direct evidence for the source rock charging reservoirs younger than the Early Miocene Peutu Formation carbonates remains limited. Gas and condensate samples from the newly discovered Rayeu CA and AOB-1 wells, along with samples from the Arun and SLS fields, are acquired and subsequently analyzed in the laboratory. Characterization and correlation of both source rock and hydrocarbon samples represent the most appropriate methodologies for evaluating their genetic relationship. Although condensates from Arun and AOB-1 exhibit retrograde phase behavior, reliable geochemical interpretations and correlations are obtained. The results indicate that gas and condensate from both Peutu carbonate and Lower Seurula sandstone are generated from a highly mature source rock dominated by algal organic matter, with only a minor contribution from higher plants. These findings suggest that multiple reservoir objectives in “B” Block, particularly in the Lhok Sukon Deep sub-basin, are most consistently correlated with a mature Bampo-type marine shale source facies.

**Keywords:** gas condensate, geochemical correlation, Lower Seurula Formation, Arun Field, North Sumatra Basin

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## 1. Introduction

The Arun and South Lhok Sukon (SLS) gas-condensate fields have been producing from the Early Miocene Peutu Formation carbonates since 1977 and 1997, respectively. After more than four decades of production, the recovery factor has reached 94% for the Arun Field and approximately 90% for the two fields combined. Despite their significant success, no major discoveries have been made in Block “B” to replace the reserves produced. Extensive exploration activities have been conducted by PT Pema Global Energi (PGE), the operator of the block since May 2021, including the acquisition and processing of 610 km<sup>2</sup> of 3D seismic data using vibroseis as the seismic source. This was followed by the drilling of two exploration wells, AOB-1 and Rayeu CA, which successfully tested gas and condensate within the Late Miocene Lower Seurula Formation sandstone reservoir intervals.

Previous studies have primarily focused on the giant Arun and South Lhok Sukon (SLS) Fields because of their substantial hydrocarbon reserves. Consequently, comprehensive petroleum system studies of stratigraphic intervals younger than the Peutu Formation remain limited. The petroleum system of the North Sumatra Basin, particularly within the North Lhok Sukon Deep sub-basin located near the Arun Field, has undergone significant reinterpretation and evolution over time. Early studies suggested that the gas and condensate accumulated within the Peutu Formation carbonate reservoirs of the Arun Field were sourced from shale intervals within the carbonate succession itself (Graves & Weegar 1973). Following revisions to the basin stratigraphic framework and nomenclature, source rocks for pre-Baong Formation reservoirs were interpreted to include shales of the Peutu, Belumai, and Bampo Formations (Kamili et al., 1976).

Subsequent geochemical analyses of well cuttings identified the Baong and Bampo Formations as significant source rock candidates. Although the Baong Formation rarely contains more than 15 wt.% total organic carbon (TOC), its thickness was considered sufficient to contribute significantly to hydrocarbon generation (Davies 1984). Using the 30-Norhopane biomarker series, Subroto et al. (1992) proposed that the oils within the Peutu Formation carbonates were derived from two potential source rocks: a mature carbonate-rich source facies and the shales of the Keutapang Formation and upper part of Baong Formation, although the maturity of these source intervals remained uncertain. Based on oil–source rock correlations using gas chromatographic and carbon isotope data, Buck and McCulloh (1994) proposed the Bampo–Peutu Petroleum System as the dominant active petroleum system in the North Sumatra Basin. In this model, the shale and marl of the Bampo Formation act as the primary source rock, while the Peutu Formation serves as both a secondary source rock and a reservoir. Within this petroleum system, the Bampo, Peutu, and Baong Formations also function as regional seals. They are overlain by the Keutapang, Seurula, and Julu Rayeu Formations, together with Quaternary deposits that constitute the basin overburden. Geochemical studies of condensates indicate that the active petroleum system is primarily associated with lacustrine source rocks containing a marine algal contribution. However, carbonate source rocks may also be present (ten Haven & Schiefelbein 1995). Fuse et al. (1996) identified the North Lhok Sukon Deep, located offshore northern Aceh, as a major hydrocarbon kitchen area, with the Bampo Formation shale acting as a Type II to Type II/III kerogen source rock.

Lacustrine facies of syn-rift sediment from the Parapat or Bruksah Formation were postulated as potential source rock based on condensate analysis done (Tambupolon et al., 2018). More recently, Ariyono et al. (2021) suggested that shallow oil accumulations in the Bireuen and East Aceh areas originated from offshore fluvio-lacustrine source rocks. In the eastern onshore area of the North Sumatra Basin, the Baong Formation has not reached maturity for hydrocarbon generation; hence, it does not cause overpressure in the formation (Sijabat et al., 2023).

No direct oil–source rock or gas–source rock correlation has been established for hydrocarbons accumulated within the Lower Seurula Formation sandstones in the North Sumatra Basin, particularly in the North Lhok Sukon Deep sub-basin. Therefore, this study aims to determine the principal source rock responsible for charging the Lower Seurula reservoirs. Based on previous studies, three hypotheses are evaluated: (1) predominant charging from the shale of the Bampo Formation; (2) contribution from the Baong Formation or other source intervals; and (3) mixed-source possibility or multi-phase charging throughout the history of the petroleum system of the basin.

This study provides new insights into the petroleum system of the North Sumatra Basin, particularly in the Lhok Sukon Deep sub-basin. It helps establish a refined exploration concept for the basin. The application of this concept is expected to reduce exploration uncertainty, improve the probability of discovery, and enhance operational efficiency, thereby strengthening the economic viability of future exploration campaigns in the “B” Block. More broadly, the results contribute to a better understanding of hydrocarbon generation and migration within the basin and may stimulate further oil and gas exploration throughout the North Sumatra Basin.

## 2. Methodology

The primary hydrocarbon dataset utilized in this study consists of one bottom-hole gas sample from the Rayeu CA well and five condensate samples collected from the Rayeu CA and AOB-1 discovery wells, together with one condensate sample each from the Arun and SLS Fields (Table 1). These samples were analyzed by the Geological and Laboratory Services Division of PT Geoservices for gas composition, gas isotop composition, liquid and gas chromatography, GC-MS (Saturate and Aromatic Fractions), stable carbon isotope analysis (Saturate and Aromatic Fractions), and pressure–volume–temperature (PVT) properties. To support the interpretation, three PVT analyses of gas samples from the Arun and Arun A-1 wells conducted by Core Laboratories were also incorporated. Geochemical data from Arun wells, including TOC, Rock-Eval pyrolysis, organic petrography and GC, were obtained from Integrated Data Services (208 cuttings and 4 core samples), Sucofindo (108 cuttings samples), and Robertson Research (1 core sample) (Table 2).

Gas and condensate characterization and correlation based on geochemical data constitute the primary methodology of this study. Hydrocarbon samples were subjected to molecular sieve isolation to improve biomarker, compositional, and stable isotope analyses. Source rock evaluation was conducted using 366 samples collected from seven wells in the Arun Field. Prior to analysis, soxhlet flushing was performed to extract the bitumen from any contaminants. The carbon isotopic composition plot from Buck and McCulloh (1994) and the sterane ternary diagram from Ariyono et al. (2021) were digitized and integrated into the analysis. In addition, the gas chromatogram of a Bampo Formation source rock extract from Buck and McCulloh (1994) is utilized for source rock-hydrocarbon correlation.

Geochemical logs were generated for each well to characterize potential source rocks with respect to organic matter quantity, quality, and thermal maturity. The interpretation of source rock parameters follows the guidelines of Merrill (1991) and Peters and Cassa (1994). Gas composition (C1, C2, C3, iC4 and nC4) and PVT data were utilized to classify the gas and condensate types, including gas wetness and molecular ratio, as in Equations 1 and 2

$$\text{Gas wetness} = \frac{(C2+C3+iC4+nC4)}{(C1+C2+C3+iC4+nC4)} \times 100\% \quad (1)$$

$$\text{Molecular ratio} = \frac{C1}{(C2+C3)} \quad (2)$$

Biomarker and carbon isotope parameters described by Peters et al. (2005) were applied to characterize the hydrocarbons, while a series of geochemical cross-plots were used to evaluate the depositional environment of the source organic matter and its maturity level, including the Methylphenanthrene Index plot to determine Equivalent Ro (Radke & Welte 1983, as cited in Peters et al., 2005) and the Tm/Ts-Moretane/Hopane ratio. The resulting geochemical characteristics, together with n-alkane gas chromatograms and tricyclic terpane distributions, were subsequently used to establish correlations between the hydrocarbons and their potential source rocks.

### 2.1 Regional geology

The North Sumatra Basin comprises a series of structural highs and depocenters. From west to east, these include the Rayeu Hinge Zone, Lhok Sukon Deep, Lhok Sukon High, Arun High, Jawa Deep, Bireuen High, and Samalanga Deep (Figure 1). The basin developed as a back-arc basin due to tectonic interaction between the Indo-

Australian and Eurasian plates from the Late Cretaceous to the Early Tertiary. The earliest stage of basin development occurred during the Late Eocene pre-rift phase, characterized by the deposition of the calcareous shales of the Meucampli Formation and the limestones and dolomites of the Tampur Formation. Rifting commenced during the Late Eocene and was marked by the development of predominantly north–south-trending horst and graben systems that controlled the initial basin architecture and sedimentation patterns. Erosion of the uplifted horsts supplied sediments to the adjacent grabens, forming localized rift valleys. These grabens were subsequently filled with syn-rift alluvial to fluvial deposits of the Bruksah Formation, representing the earliest clastic sediments deposited during basin evolution.

The Late Oligocene–Early Miocene sag phase was characterized by regional subsidence and a progressive increase in water depth, leading to the inundation of the earlier rift topography. Anoxic to dysoxic depositional conditions facilitated the accumulation of the organic-rich shales of the Bampo Formation, marking the initiation of post-rift sedimentation. Continued marine transgression during the Early to Middle Miocene promoted extensive carbonate platform development of the Peutu Formation over structural highs. Deposition of the Baong Formation shale during the Middle Miocene maximum flooding event subsequently terminated carbonate growth and provided an effective regional seal. Together, the Bampo, Peutu, and Baong Formations constitute the key source, reservoir, and seal elements of the North Sumatra Basin petroleum system.

Middle Miocene wrench tectonism marked the transition from the post-rift to the regressive phase of basin evolution and was accompanied by uplift of the Barisan Mountain Range. This tectonic reorganization promoted the deposition of the upper Baong Formation and the subsequent accumulation of fluvio-deltaic sediments of the Keutapang and Seurula Formations. Continued compressional deformation and associated volcanism during the Pliocene–Pleistocene reactivated earlier extensional faults, resulting in structural inversion and the development of numerous hydrocarbon traps. These tectonic processes were fundamental to the migration, entrapment, and preservation of hydrocarbons within the North Sumatra Basin.

Table 1.  
Gas and Condensate dataset from Rayeu CA, AOB-1,  
Arun A-1 wells and Arun-SLS Fields production facilities.

| Well/<br>Field | Sample<br>Type | Sample<br>Quantity | Data<br>Acquisition   | Analysis     |                         |                  |               |           |                             |     | Lab./<br>Report/<br>Source |
|----------------|----------------|--------------------|-----------------------|--------------|-------------------------|------------------|---------------|-----------|-----------------------------|-----|----------------------------|
|                |                |                    |                       | Gas<br>Comp. | Gas<br>Isotope<br>Comp. | Liquid<br>Chrom. | Gas<br>Chrom. | GC-<br>MS | Stable<br>Carbon<br>Isotope | PVT |                            |
| Rayeu<br>CA    | Gas            | 1                  | Bottom Hole<br>Sample | v            | v                       |                  |               |           |                             | v   | Geoservices<br>2024        |
|                | Condensate     | 1                  | Separator             |              |                         | v                | v             | v         | v                           | v   |                            |
| AOB-<br>1      | Condensate     | 2                  | Separator             |              |                         | v                | v             | v         | v                           | v   |                            |
| SLS            | Condensate     | 1                  | Separator             |              |                         | v                | v             | v         | v                           |     |                            |
| Arun           | Condensate     | 1                  | Separator             |              |                         | v                |               |           |                             |     | Corelab<br>1972            |
|                | Gas            | 1                  | Separator             | v            |                         |                  |               |           |                             | v   |                            |
| Arun<br>A-1    | Gas            | 2                  | Separator             | v            |                         |                  |               |           |                             | v   | Buck &<br>McCulloh<br>1994 |
| Arun           | Condensate     | 10                 | n/a                   |              |                         |                  |               |           | v                           |     |                            |

### 3. Result and Discussion

The majority of source rock data were derived from wells in the Arun Field, located on the Arun High. Among the wells evaluated, only East Arun A-1 was drilled within the Lhok Sukon Deep (Figure 3).

### 3.1 Source rock determination

Geochemical logs were constructed using data from four wells: Arun A-1, Arun A-7, Arun A-13, and East Arun A-1 (Figure 3). The discovery well of the Arun Field, Arun A-1, provided 13 cutting samples for analysis. Among these, only one black-shale sample from the Bampo Formation exhibited good source rock potential, with a TOC of 2.16 wt.% and a Tmax value of 480°C. Most samples from Arun A-7 were obtained from the Baong Formation shale and exhibited poor source rock quality, with all TOC values below 0.5 wt.%. Rock-Eval pyrolysis (REP) results from extracted samples of Arun A-13 suggest source rock potential within the lower Keutapang Formation, Baong Formation, and Bampo Formation intervals. East Arun A-1 provided 108 cutting samples spanning the Upper Seurula Formation at approximately 5,000 ft MD to the Bruksah Formation at 15,700 ft MD. Unlike the other wells, East Arun A-1 is located within the Lhok Sukon Deep and offers valuable insight into the deeper portion of the basin. Although several stratigraphic intervals exhibit source rock potential, geochemical data indicate that source rocks encountered in the Arun High are generally immature. This interpretation is supported by both vitrinite reflectance measurements and Rock-Eval Tmax values, which suggest insufficient thermal maturity for significant hydrocarbon generation.

Table 2.  
Source rock dataset from Arun wells.

| Well          | Formation     | Sample Type     | Depth (ft MD)       | Sample Quantity |      |     | Analysis |    |    |      |    |          |    |        | Lab./ Report/ Source                              |
|---------------|---------------|-----------------|---------------------|-----------------|------|-----|----------|----|----|------|----|----------|----|--------|---------------------------------------------------|
|               |               |                 |                     | Cuttings        | Core | TOC | S1       | S2 | S3 | Tmax | Ro | Bitu men | GC | Isotop |                                                   |
| Arun A-1      | Peutu         | Cuttings        | 10,525              | 1               | -    | v   |          |    |    |      |    |          |    |        | Study Report Technology Research Center JNOC 1998 |
|               | Bampo         | Cuttings        | 10,550 - 10,810     | 8               | -    | v   | v        | v  | v  | v    |    |          |    |        |                                                   |
|               | Basement      | Cuttings        | 10,847.5 - 10,883   | 4               | -    | v   | v        | v  | v  | v    |    |          |    |        |                                                   |
| Arun A-6      | Peutu         | Core            | 10,067              | -               | 1    | v   | v        | v  |    | v    |    |          |    |        |                                                   |
| Arun A-7      | Julu Rayeu    | Cuttings        | 1,000 - 3,070       | 3               | -    |     |          |    |    |      | v  |          |    |        |                                                   |
|               | Seurula       | Cuttings        | 4,000 - 4,030       | 1               | -    |     |          |    |    |      | v  |          |    |        |                                                   |
|               | Keutapang     | Cuttings        | 5,140 - 8,715       | 16              | -    | v   | v        | v  | v  | v    | v  |          |    |        |                                                   |
|               | Baong         | Cuttings        | 8,745 - 10,142.5    | 40              | -    | v   | v        | v  | v  | v    | v  |          |    |        |                                                   |
|               | Peutu         | Core & Cuttings | 10,173              | 1               | 1    | v   | v        | v  |    | v    |    |          |    |        |                                                   |
| Arun A-13     | Keutapang     | Cuttings        | 7,950 - 8,550       | 19              | -    | v   |          |    |    |      | v  |          |    |        | Well Report (Sucofindo) 1990                      |
|               | Baong         | Core & Cuttings | 8,570 - 10,450      | 88              | 1    | v   | v        | v  | v  | v    | v  |          |    |        |                                                   |
|               | Peutu         | Cuttings        | 10,470 - 10,490     | 2               | -    | v   |          |    |    |      |    |          |    |        |                                                   |
|               | Bampo         | Cuttings        | 10,567.5 - 11,076.5 | 25              | -    | v   | v        | v  | v  | v    | v  |          |    |        |                                                   |
| Arun A-54     | Baong         | Core            | 10,347.50           | -               | 1    | v   | v        | v  |    | v    |    |          |    |        |                                                   |
| Arun East A-1 | Upper Seurula | Cuttings        | 5,000 - 5,900       | 10              | -    | v   | v        | v  | v  | v    | v  |          |    |        |                                                   |
|               | Lower Seurula | Cuttings        | 6,000 - 6,800       | 9               | -    | v   | v        | v  | v  | v    | v  |          |    |        |                                                   |
|               | Keutapang     | Cuttings        | 6,900 - 8,700       | 19              | -    | v   | v        | v  | v  | v    | v  | v        | v  |        |                                                   |
|               | Baong         | Cuttings        | 8,800 - 12,800      | 41              | -    | v   | v        | v  | v  | v    | v  | v        | v  |        |                                                   |
|               | Peutu         | Cuttings        | 12,900 - 13,400     | 6               | -    | v   | v        | v  | v  | v    | v  | v        | v  |        |                                                   |
|               | Bampo         | Cuttings        | 13,500 - 15,300     | 19              | -    | v   | v        | v  | v  | v    | v  | v        | v  |        |                                                   |
|               | Bruksah       | Cuttings        | 15,400 - 15,700     | 4               | -    | v   | v        | v  | v  | v    | v  | v        | v  |        |                                                   |
| Arun B-1      | Pre-Tertiary  | Core            | 10,335              | -               | 1    | v   |          |    |    |      | v  |          |    |        | Well Report (Robertson Research Singapore) 1980   |
| unknown       | Baong         | unknown         | n/a                 | 20              | -    |     |          |    |    |      |    |          |    | v      | Buck & McCulloh 1994 (Digitized)                  |
|               | Bampo         | unknown         |                     | 20              | -    |     |          |    |    |      |    |          | v  | v      |                                                   |
|               | Baong         | Outcrop         |                     | 3               | -    |     |          |    |    |      |    |          |    | v      |                                                   |
|               | Bampo         | Outcrop         |                     | 2               | -    |     |          |    |    |      |    |          |    | v      |                                                   |

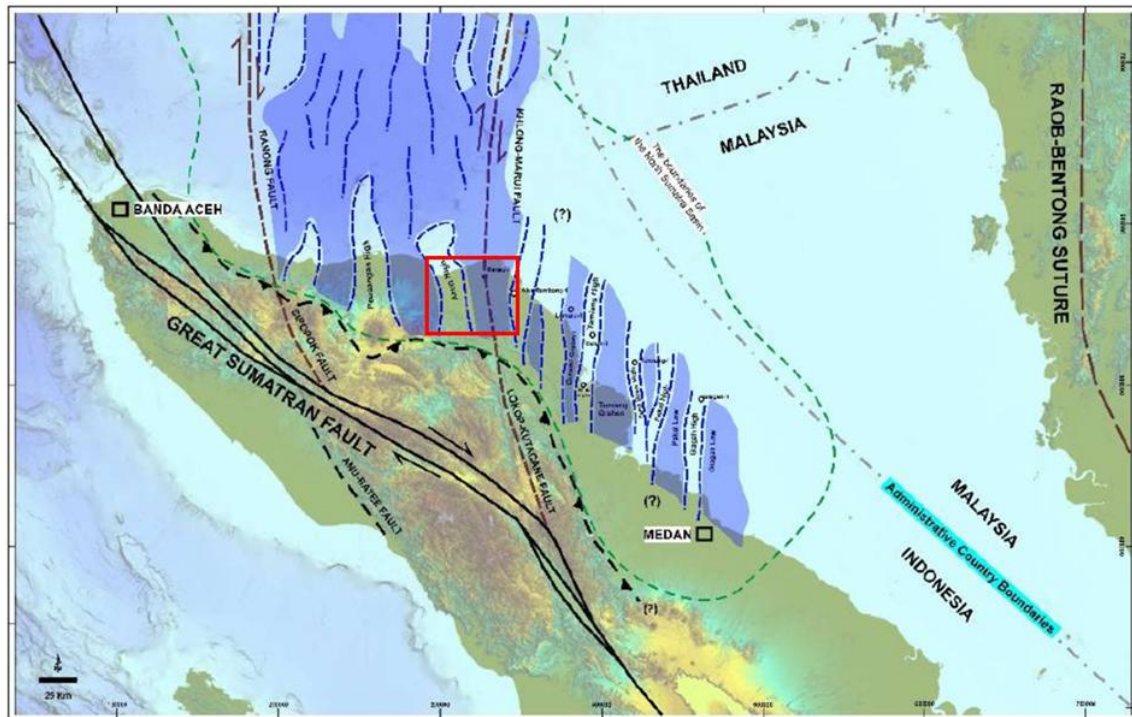


Figure 1.  
Study area (red square) located in the North Sumatra Basin, Lhok Sukon Deep in particular (Pasaribu et al., 2025).

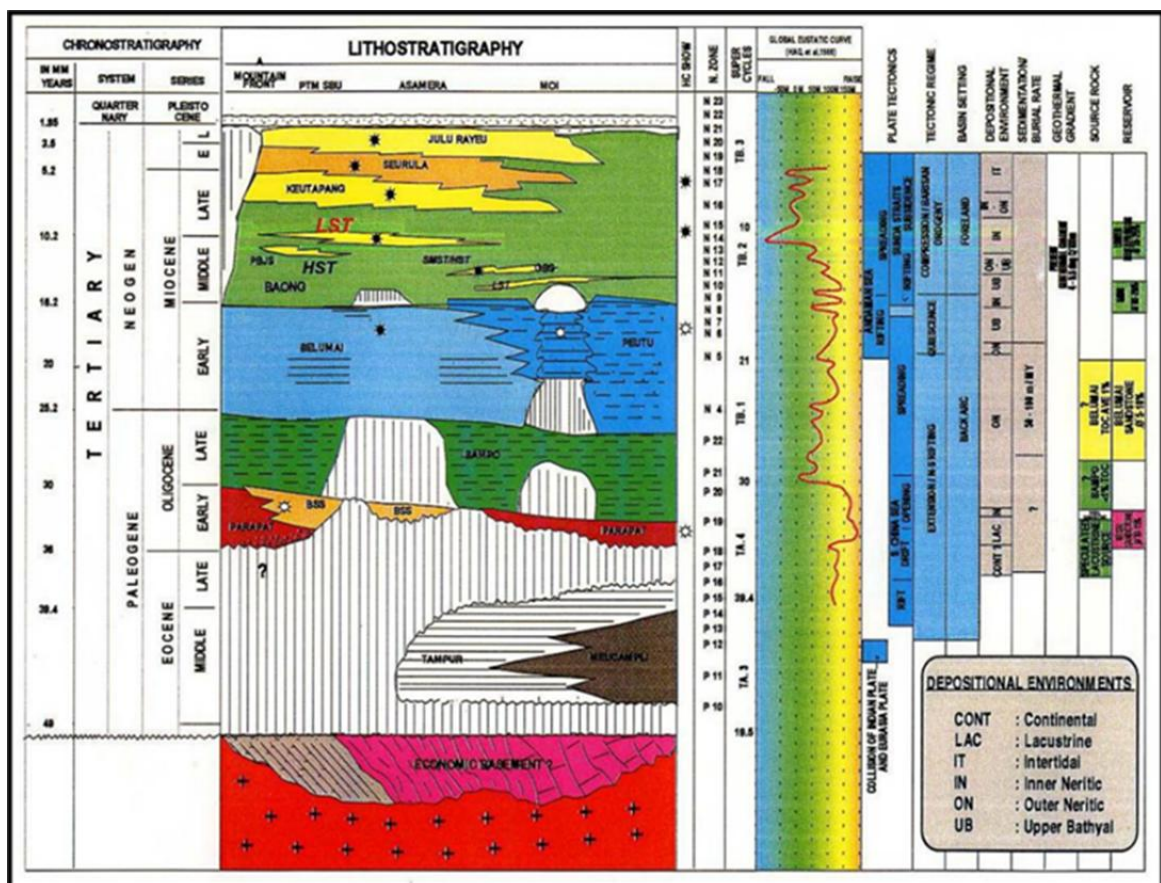


Figure 2.  
Tectonostratigraphic chart of North Sumatra Basin (Riadhy et al., 1998).

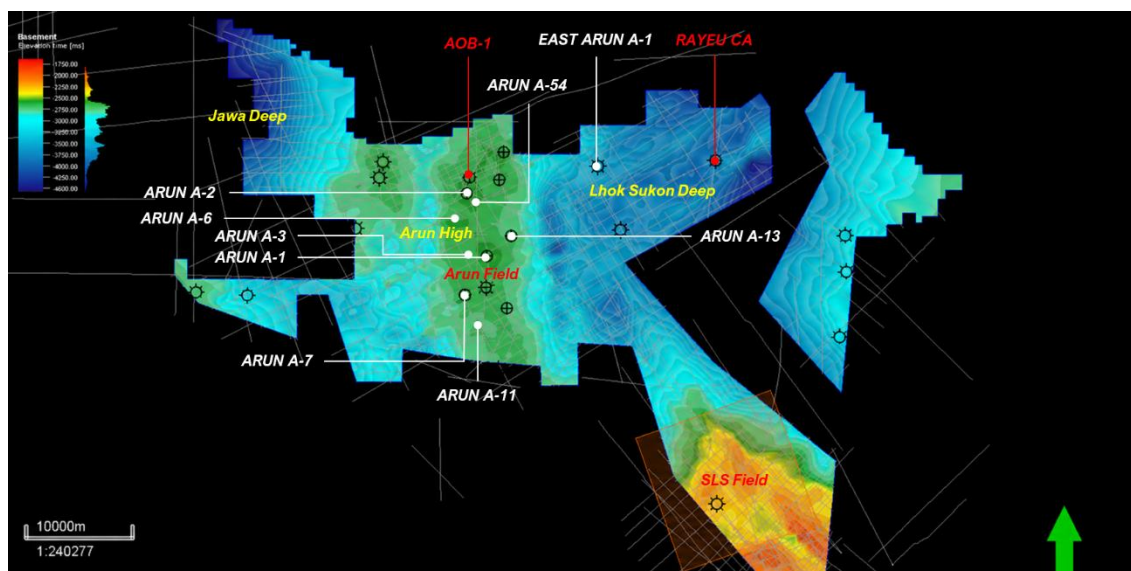


Figure 3.  
Well data distribution used for the study,  
showing that most of the data are located on the Arun High.

The geochemical analyses conducted on samples from East Arun A-1 are considerably more comprehensive than those performed on the other wells. In addition to total organic carbon (TOC) measurements, Rock-Eval pyrolysis (REP), and organic petrographic analyses, gas chromatography was performed (Figure 4). Quantitatively, source rock intervals from the Keutapang Formation down to the Bruksah Formation exhibit fair to good source rock potential. Based on the vitrinite reflectance trend across the entire stratigraphic section, thermal maturity reaches the onset of the oil-generation window (approximately 0.6 %Ro) within the lower part of the Baong Formation. This observation is significant because East Arun A-1 is the only well in the dataset located within the Lhok Sukon Deep sub-basin. The Van Krevelen diagram indicates that the potential source rock intervals are dominated by Type II and Type II/III kerogen (Figure 5), suggesting a predominantly oil- and gas-prone organic matter assemblage. Combined with the observed maturity trend, these results demonstrate that the deeper stratigraphic succession in the Lhok Sukon Deep possesses favorable source rock characteristics and has begun to enter the hydrocarbon generation window. Gas chromatograms of rock extracts from Baong and Bampo Formations in the East Arun A-1 wells show peak n-alkane distributions between  $nC_{17}$ - $nC_{19}$ . Corresponding Pristane/Phytane and Pristane/ $nC_{17}$  ratios indicate the predominance of organic matter deposited under anoxic marine conditions, with a minor contribution from higher terrestrial plant material (Figure 6).

### 3.2 Gas characterization

Only one gas sample was collected from a bottom-hole sample in the Rayeu CA well, after a mechanical problem was encountered during the drilling of the AOB-1 well. The sample was submitted for gas composition and stable carbon isotope analyses. The gas contains a high proportion of hydrocarbons (84.43 mol.%), dominated by methane (70.17 mol.%). Its relatively high gas wetness value (14.9%) indicates that it is a wet gas (Table 3). A low  $C1/(C2+C3)$  molecular ratio further suggests a thermogenic origin. In addition, the absence of hydrogen sulfide classifies the gas as a sweet natural gas. The stable carbon isotope composition shows an approximately 20‰ difference between methane and ethane ( $\delta^{13}C_2 - \delta^{13}C_1$ ), suggesting a minor contribution of biogenic methane mixed with predominantly thermogenic gas (Figure 6). Furthermore, the gas maturity was evaluated using the co-genetic ethane–propane relationship proposed by Faber (in Whiticar 1994). Application of this method suggests that the gas within the Lower Seurula sandstone reservoir from the Rayeu CA well was generated from a highly mature source rock at an advanced stage of thermal maturity and is predominantly thermogenic, corresponding to an equivalent vitrinite reflectance of approximately 2.0 %Ro.

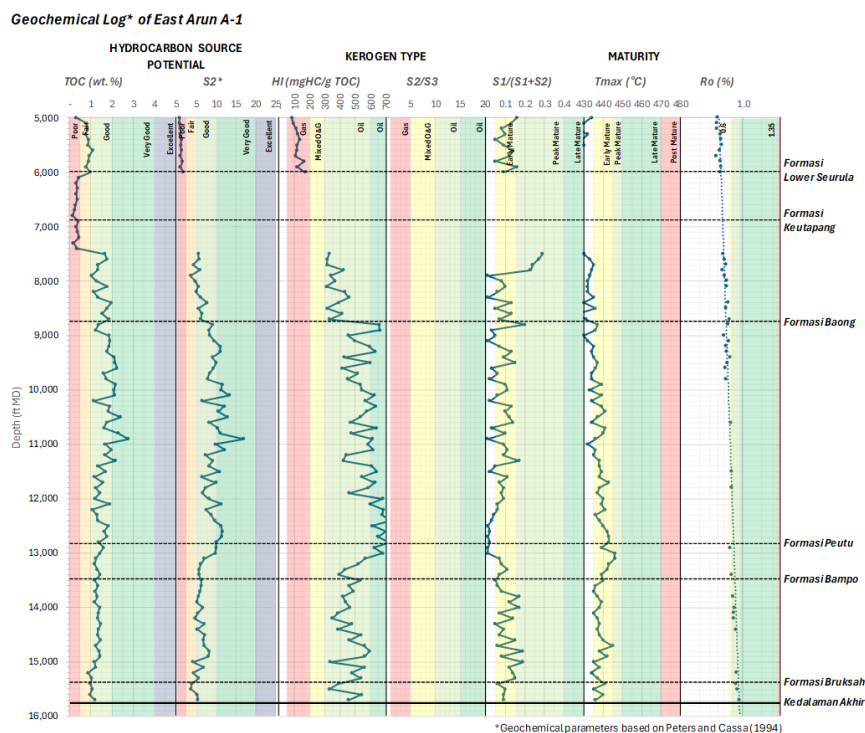


Figure 4.

Geochemical Log of East Arun A-1 well, showing potential source rock intervals based on organic matter quantity, quality and thermal maturity. The identified candidates include the Baong, Peutu, Bampo and Bruksah formations.

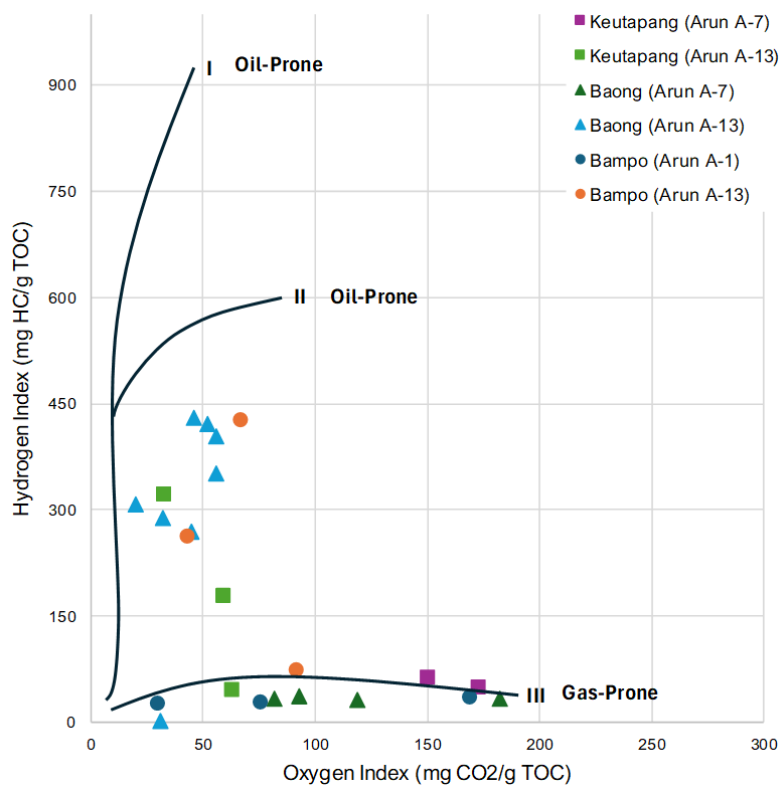


Figure 5.

van Krevelen Diagram of cuttings samples from Arun wells, exhibit mixed type II and type III kerogen within the identified source rock candidates.

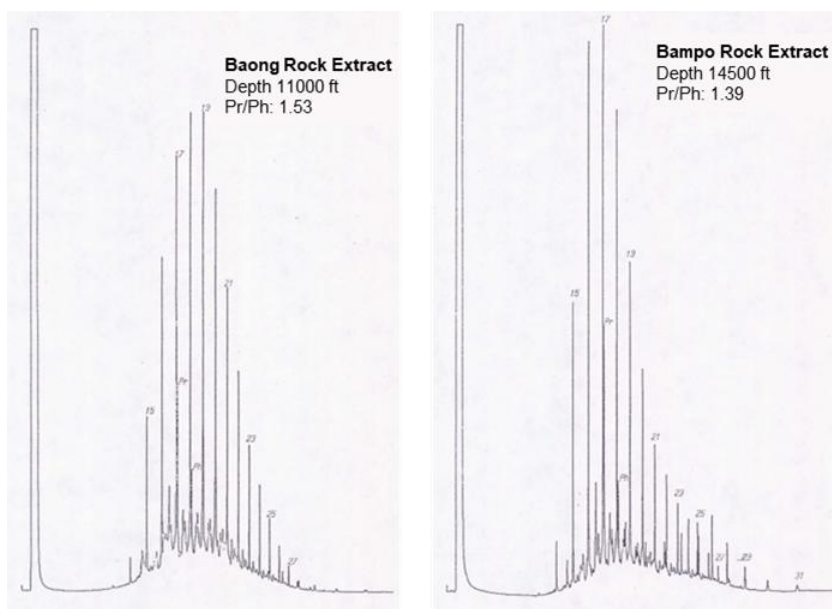


Figure 6.  
n-Alkane distributions from Rock Extracts of East Arun A-1 well.

Table 3.  
Gas Composition from Rayeu CA well.

|                 | Component                                         | mol.% | Normalized<br>% HC | $\delta^{13}\text{C}$<br>(per mille) |
|-----------------|---------------------------------------------------|-------|--------------------|--------------------------------------|
| Non-Hydrocarbon | He                                                |       |                    |                                      |
|                 | H <sub>2</sub>                                    |       |                    |                                      |
|                 | Ar                                                | 0.13  |                    |                                      |
|                 | H <sub>2</sub> S                                  |       |                    |                                      |
|                 | O <sub>2</sub>                                    | 3.11  |                    |                                      |
|                 | CO <sub>2</sub>                                   | 0.89  |                    | -7.91                                |
|                 | N <sub>2</sub>                                    | 11.44 |                    |                                      |
| Hydrocarbon     | C <sub>1</sub>                                    | 70.17 | 83.10              | -45.51                               |
|                 | C <sub>2</sub>                                    | 6.95  | 8.23               | -25.12                               |
|                 | C <sub>3</sub>                                    | 3.29  | 3.90               | -22.94                               |
|                 | iC <sub>4</sub>                                   | 0.96  | 1.14               |                                      |
|                 | nC <sub>4</sub>                                   | 1.09  | 1.29               |                                      |
|                 | iC <sub>5</sub>                                   | 0.61  | 0.72               |                                      |
|                 | nC <sub>5</sub>                                   | 0.36  | 0.43               |                                      |
|                 | C <sub>6+</sub>                                   | 1.01  | 1.20               |                                      |
|                 | Total Gas (HC)                                    | 84.44 | 100.00             |                                      |
|                 | C <sub>1</sub> /(C <sub>2</sub> +C <sub>3</sub> ) | 6.85  | 6.85               |                                      |
|                 | Gas wetness                                       | 14.90 | 14.90              |                                      |

### 3.3 Condensate characterization

An internal Arun Field fluid model report (1991) classified the Arun reservoir fluid as a retrograde condensate based on constant-volume depletion (CVD) tests performed on PVT samples from Arun A-1 and the Cluster II Production Facility (Figure 8). Two samples from Arun A-1 were analyzed at different reservoir depths. The shallower sample (9,953–9,958 ft MD) exhibited a retrograde dew-point pressure of 4,426 psig at 348°F, whereas the deeper sample (10,410–10,415 ft MD) showed a dew-point pressure of 4,348 psig at 368°F. Both samples were obtained from a reservoir with a static pressure of 7,085 psig, and the maximum liquid dropout observed during depletion was approximately 0.7% of the original fluid volume. A surface sample collected from the

Cluster II Production Facility also displayed retrograde condensate behavior, with a dew-point pressure of 4,411 psig at 352°F and a maximum liquid dropout of 1.13% of the original fluid volume. Similar results were obtained from two AOB-1 reservoir fluid samples, designated Zone 1 and Zone 2. The Zone 1 sample exhibited a dew-point pressure of 3,044 psig at 224°F and a maximum liquid dropout of 14.1%, whereas the deeper Zone 2 sample showed a dew-point pressure of 4,180 psig at 225°F and a maximum liquid dropout of 15.8%. The occurrence of retrograde condensates has important implications for this study. On the one hand, these fluids retain sufficient biomarker compounds to support reliable geochemical characterization and correlation. On the other hand, preferential partitioning of lighter hydrocarbons into the gas phase alters the distribution of n-alkanes and isoprenoids, affecting parameters such as the carbon preference index (CPI) and absolute chromatographic peak abundances. Retrograde condensates are commonly associated with hydrocarbons generated from Type II or Type II/III kerogen at advanced thermal maturity.

Table 4.  
Bulk composition, biomarker and Stable Carbon Isotope of Condensate from Rayeu CA, AOB-1, Arun and SLS Fields.

| Sample                        | Arun Cond. | AOB-1 Cond.        | Rayeu CA Cond.     | SLS Cond.  |
|-------------------------------|------------|--------------------|--------------------|------------|
| Reservoir                     | Peutu Lst. | Lower Seurula Sst. | Lower Seurula Sst. | Peutu Lst. |
| <i>HC. Compositions (%)</i> : |            |                    |                    |            |
| Saturates                     | 84.56      | 80.50              | 95.77              | 83.91      |
| Aromatics                     | 15.05      | 17.88              | 3.91               | 15.58      |
| NSOs                          | 0.37       | 1.50               | 0.30               | 0.49       |
| Asph.                         | 0.02       | 0.12               | 0.02               | 0.02       |
| <i>Biomarker</i>              |            |                    |                    |            |
| Pristane                      | 53,879     | 67,898             | 14,979             | 77,865     |
| Phytane                       | 16,344     | 24,906             | 4,666              | 35,472     |
| Pristane / Phytane            | 3.30       | 2.73               | 3.21               | 2.20       |
| Pristane / nC17               | 0.59       | 0.75               | 0.58               | 0.30       |
| Phytane / nC18                | 0.33       | 0.34               | 0.30               | 0.18       |
| <i>n-Alkane</i>               |            |                    |                    |            |
| A. Total C25 - C33 Odd        | 1,192      | 14,686             | 761                | 8,741      |
| B. Total C24 - C32 Even       | 2,247      | 23,182             | 1,193              | 18,065     |
| C. Total C26 - C34 Even       | 538        | 9,280              | 392                | 3,863      |
| CPI 1                         | 1.37       | 1.11               | 1.29               | 1.37       |
| D. Total C23 - C27 Odd        | 4,298      | 33,119             | 2,034              | 35,547     |
| E. Total C25 - C29 Odd        | 1,192      | 14,160             | 761                | 8,741      |
| F. Total C24 - C28 Even       | 2,247      | 22,021             | 1,193              | 17,927     |
| CPI 2                         | 1.22       | 1.07               | 1.17               | 1.24       |
| <i>Terpane m/z 191</i>        | Peak ID    |                    |                    |            |
| Tricyclic terpane             | A          | 1,450              | 322                | 285        |
| Tricyclic terpane             | B          | 370                | 226                | 156        |
| Tricyclic terpane             | C          | 415                | 460                | 160        |
| Tricyclic terpane             | D          | 298                | 456                | 247        |
| Tricyclic terpane             | E          | 43                 | 227                | 95         |
| Tricyclic terpane             | F          | 131                | 352                | 167        |
| Tricyclic terpane             | G          | 56                 | 232                | 87         |
| Tricyclic terpane             | H          | 19                 | 70                 | 27         |
| Tm/Ts                         | 0.84       | 0.96               | 1.43               | 0.96       |
| Ts/(Ts+Tm)                    | 0.54       | 0.51               | 0.41               | 0.51       |
| C30 Mor./C30 Hopane           | -          | 0.17               | 0.20               | -          |
| <i>Sterane m/z 217</i>        |            |                    |                    |            |
| ααα C27 Steranes              | 41.86      | 47.20              | 53.23              | 33.33      |
| ααα C28 Steranes              | 20.93      | 21.01              | 19.77              | 28.57      |
| ααα C29 Steranes              | 37.21      | 31.79              | 27.00              | 38.10      |
| <i>Stable Carbon Isotope</i>  |            |                    |                    |            |
| Saturates                     | -24.50     | -23.20             | -24.80             | -20.80     |
| Aromatics                     | -22.30     | -23.60             | -24.70             | -18.60     |

Bulk composition analysis indicates that condensate samples from the AOB-1 and Rayeu CA wells, as well as from the Arun and SLS Fields, contain 80.5–95.8% saturated hydrocarbons, 3.9–17.9% aromatic hydrocarbons, and only 0.3–1.6% NSO and asphaltene compounds. This composition demonstrates that the hydrocarbons accumulated within the Arun and SLS Fields and the AOB and Rayeu structures are predominantly paraffinic condensates. In contrast, the bitumen sample obtained from the East Arun A-1 well exhibits a lower level of thermal maturity than the condensate samples from the Arun and SLS Fields and the Rayeu CA and AOB-1 wells (Figure 9). This maturity difference is reflected in the compositional characteristics of the bitumen and suggests that it represents a less thermally evolved hydrocarbon phase.

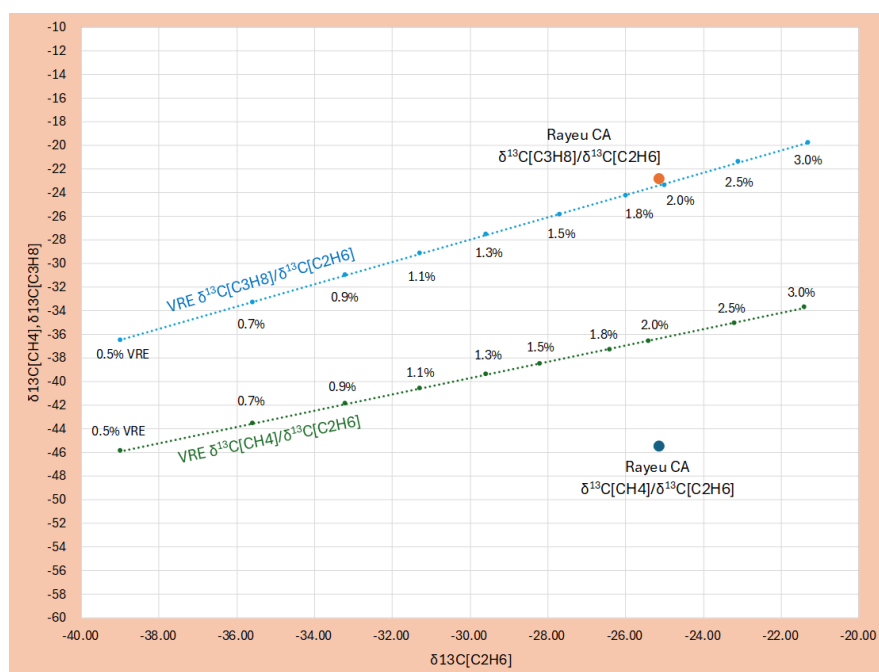


Figure 7.

The Whiticar's cross-plot (1994) of the stable carbon isotopic compositions of Ethane to Methane and Ethane from Rayeu CA well, indicating that the gas was generated from a highly mature source rock.

The n-alkane distributions of the condensate samples suggest a predominantly marine source of organic matter with a minor terrestrial input (Figure 10). Pristane/phytane (Pr/Ph) ratios indicate deposition under anoxic to suboxic redox conditions. In addition, relatively low Ph/nC<sub>18</sub> ratios coupled with elevated Pr/Ph values imply a shale-dominated source facies. Collectively, these geochemical parameters point to hydrocarbons generated from marine algal organic matter deposited in a reducing environment, consistent with a marine shale source rock.

The sterane ternary diagram indicates that hydrocarbons from the AOB-1 and Rayeu CA wells, produced from the Lower Seurula Formation sandstone reservoirs, were generated from a source rock dominated by marine organic matter. This interpretation is consistent with the characteristics of hydrocarbons produced from the Peutu Formation carbonate reservoirs of the Arun Field, as well as those from the Seurula Formation sandstone reservoirs in the Bireuen area. Furthermore, condensate samples from the South Lhok Sukon (SLS) Field and the East Aceh area also exhibit geochemical signatures indicative of predominantly marine organic matter, with a minor contribution from terrestrial input (Figure 11). The similarity in sterane distributions among the Arun, SLS, AOB, Rayeu, Bireuen, and East Aceh hydrocarbon samples suggests a common genetic affinity and supports the interpretation that these accumulations were derived from a similar source rock facies. The deviation observed for Arun A-1/ST in the sterane ternary diagram does not necessarily indicate a different depositional setting; rather, it likely reflects increased input from higher plants into the main environment. The predominance of marine organic matter, coupled with limited terrestrial influence, is consistent with a marine shale source rock deposited under relatively reducing conditions.

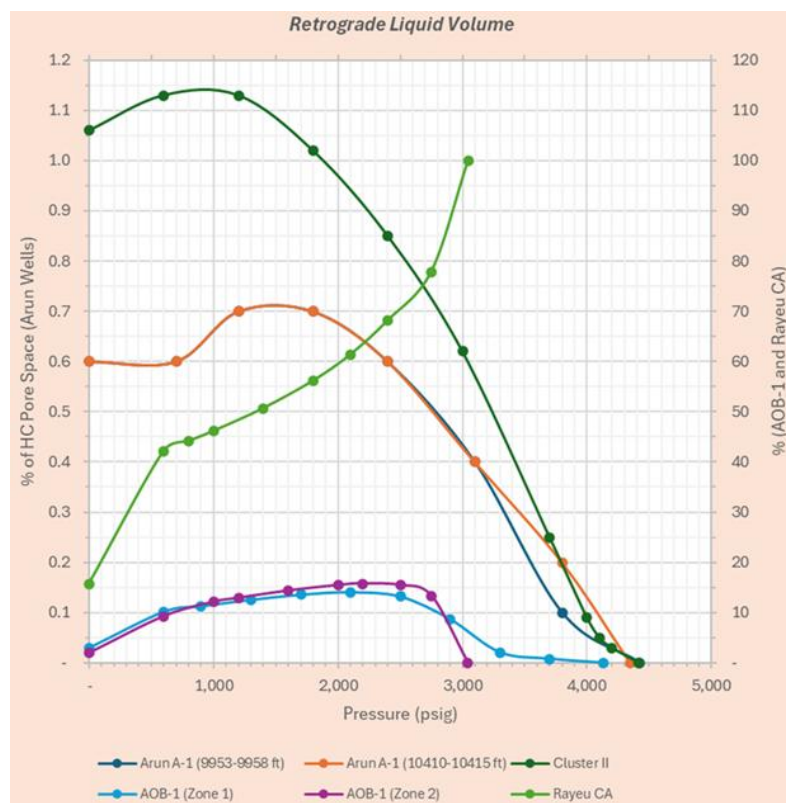


Figure 8.

Cross-plot of retrograde liquid volume based on constant volume depletion from PVT gas and condensate from Arun A-1, Cluster II, AOB-1 and Rayeu CA. The retrograde condensates represent type II or II/III kerogen of highly mature source rock.

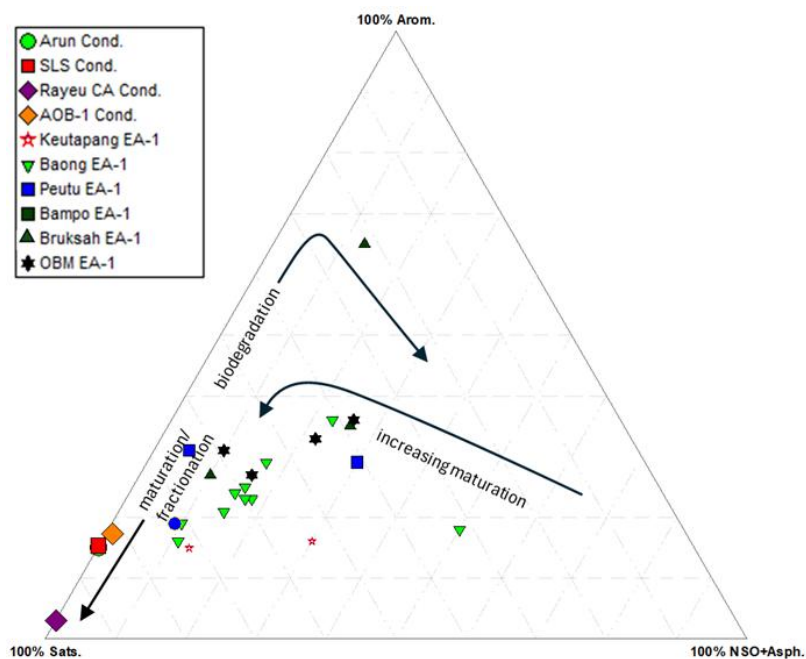


Figure 9.

Ternary diagram of SANA compositions of condensates from the AOB-1, Rayeu CA and Arun-SLS wells, showing the condensates are more thermally mature than the identified source rock candidates. This relationship suggests generation from deeper and more mature source rock intervals within the basin.

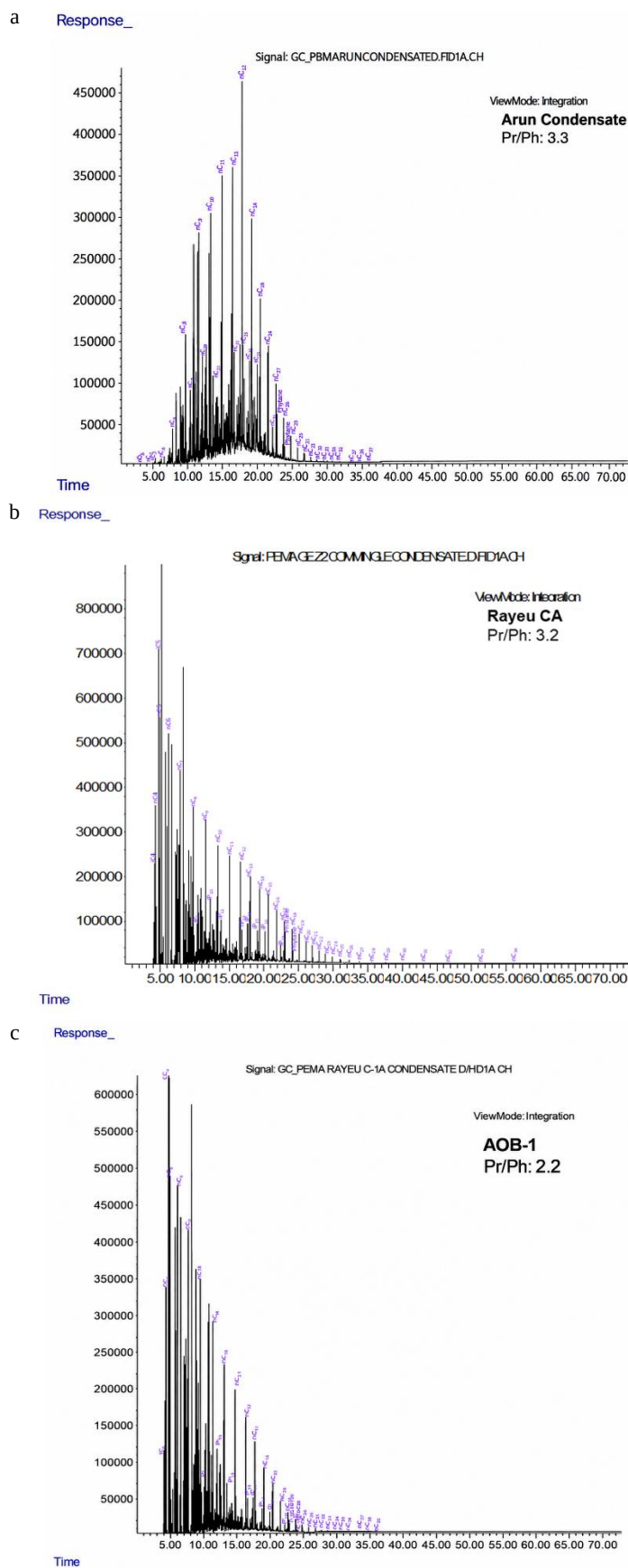


Figure 10.  
n-Alkane distributions from condensate samples of AOB-1 (a), Rayeu CA (b) and Arun Field (c),  
showing a predominance of marine-derived organic matter.

Based on the GC-MS  $m/z$  191 tricyclic terpene distribution, the condensate samples are characterized by a dominant  $C_{30}(H)$ -hopane peak, with  $C_{29}$  hopane occurring in lower abundance than  $C_{30}$  hopane. This biomarker distribution is characteristic of hydrocarbons derived from a siliciclastic (clastic) source rock rather than a carbonate source rock. In addition, the tricyclic terpene distributions suggest that the hydrocarbons were generated from a mixed assemblage of organic matter, predominantly algal, with a minor contribution from terrestrial higher plants (Figure 12). The consistency of these biomarker signatures among the analyzed condensate samples indicates a common source facies. It supports the interpretation that the hydrocarbons were derived from a marine shale source rock containing mixed marine and terrestrial organic matter input. This interpretation agrees with the sterane distributions and other geochemical parameters, which collectively point to a predominantly marine depositional environment with minor terrestrial influence.

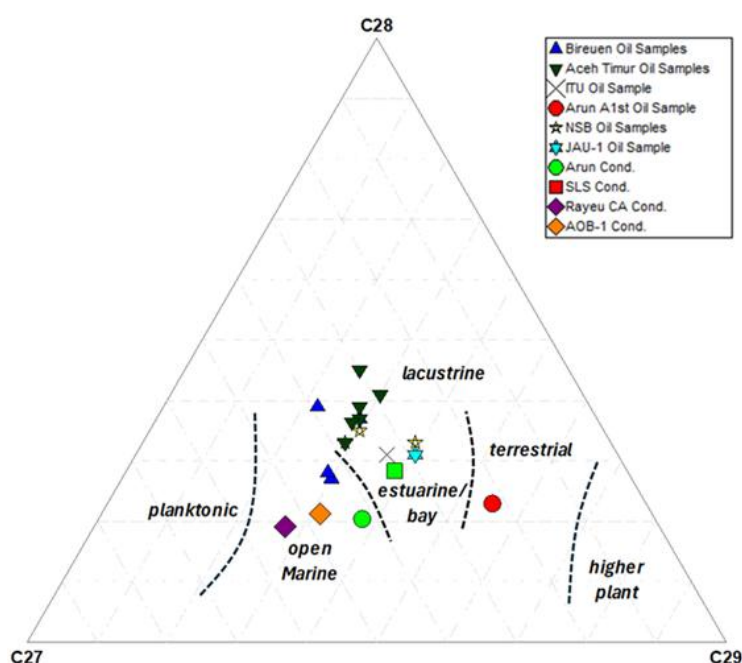


Figure 11.

Sterane ternary diagram showing the dominance of marine organic matter influence in the hydrocarbon samples analyzed in this study, integrated with the dataset of Ariyono et. al. (2021).

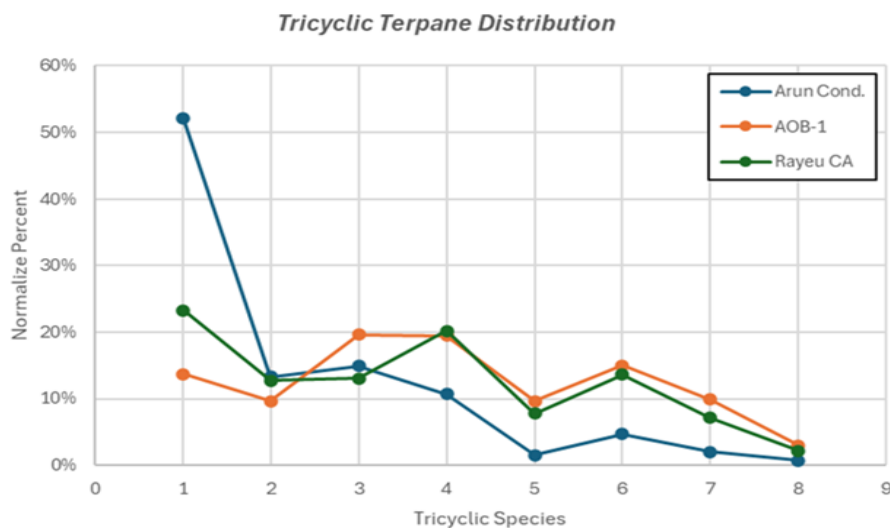


Figure 12.

Tricyclic Terpene distribution for A/L peaks ( $m/z$  191), indicating a mixture of algal and higher plant organic matter input.

Maturity assessment based on triterpane biomarker ratios indicates that the Rayeu CA condensate is within the early mature stage, whereas the AOB-1 condensate has reached peak thermal maturity. This interpretation is consistent with results derived from the methylphenanthrene maturity index (MPI), which generally indicate that the analyzed condensate samples were generated within the peak oil-generation window. The corresponding equivalent vitrinite reflectance (EqVR) values range from 0.88 to 1.08 %Ro, further supporting a maturity level characteristic of peak hydrocarbon generation (Figure 13). Although minor variations in maturity exist between the samples, the overall biomarker and aromatic hydrocarbon maturity parameters indicate that the condensates were generated from thermally mature source rocks. These results are consistent with the high maturity inferred from gas geochemistry and support the interpretation that the hydrocarbons were derived from a mature petroleum system capable of generating both gas and condensate.

### 3.4 Gas-condensate to source rock correlation

The condensate samples from the Arun Field, AOB-1, and Rayeu CA wells exhibit a unimodal n-alkane distribution characterized by the predominance of nC<sub>15</sub>, nC<sub>17</sub>, and nC<sub>19</sub> over nC<sub>27</sub>, nC<sub>29</sub>, and nC<sub>31</sub>, together with relatively low abundances of odd-carbon-numbered homologs in the nC<sub>23</sub>–nC<sub>31</sub> range. This distribution indicates a dominant contribution from marine organic matter, with only a minor input from terrestrial sources. Although the n-alkane envelope patterns of the Baong and Bampo Formation extracts are broadly similar, they differ in the position of their dominant peaks: the Baong Formation extract peaks at nC<sub>19</sub>, whereas the Bampo Formation extract peaks at nC<sub>17</sub>. Furthermore, the gradual decline in the abundance of nC<sub>20</sub>+ hydrocarbons observed in the condensate samples closely resembles the corresponding distribution pattern in extracts from the Bampo Formation, suggesting a stronger genetic affinity with this source rock (Figure 10).

Compared with the condensate samples, the source rock extracts contain a greater proportion of lighter hydrocarbons and generally exhibit lower Pr/Ph ratios (Figure 14). These differences, including a lower source rock maturity parameter of 0.6 %Ro from East Arun A-1 compared to the condensate and gas maturity levels, indicate that the condensates are thermally more mature than the analyzed source rock extracts and, therefore, are unlikely to have been generated locally from the analyzed intervals. Instead, the hydrocarbons were most likely expelled from a more deeply buried and thermally mature source kitchen (Figures 3 and 15). This interpretation suggests that the principal hydrocarbon kitchen is in the deeper part of the basin, north of the present-day Aceh coastline, within the Lhok Sukon Deep sub-basin. Hydrocarbons generated from the Bampo Formation shale are interpreted to have migrated vertically through the inverted bounding faults of the Lhok Sukon Deep and associated propagating fault systems. Subsequent lateral migration occurred along permeable Lower Seurula sandstone carrier beds, resulting in hydrocarbon charge and accumulation within the Lower Seurula reservoirs.

Overall, the condensate and source rock samples appear to have been derived from a source rock that contains predominantly marine organic matter with a minor terrestrial contribution. The Arun condensate sample is isotopically slightly heavier than its nearest Bampo Formation extract sample, with a difference of less than 1‰. In contrast, the isotopic difference between the Arun condensate and Baong Formation extract samples ranges from approximately 2‰ to 4‰. Similarly, the AOB-1 condensate sample is isotopically slightly heavier than the closest Bampo Formation extract sample, with a difference of less than 1‰. In contrast, its isotopic separation from the Baong Formation extract samples exceeds 1‰. The Rayeu CA condensate sample is isotopically slightly heavier than both the Baong and Bampo Formation extract populations. However, the isotopic composition of the saturated hydrocarbon fraction shows the closest affinity with the Bampo Formation extracts rather than those from the Baong Formation (Figure 16). These isotopic relationships indicate a stronger genetic correlation between the condensates and the Bampo Formation source rock extracts. The consistently small isotopic differences observed between the Arun, AOB-1, Rayeu CA condensates, and the Bampo Formation extracts support the interpretation that the Late Oligocene Bampo Formation shale represents the principal source rock for hydrocarbons accumulated in the Arun, SLS, AOB, and Rayeu structures. Conversely, the larger isotopic offsets relative to the Baong Formation extracts suggest that the Baong Formation played a less significant role in charging these accumulations.

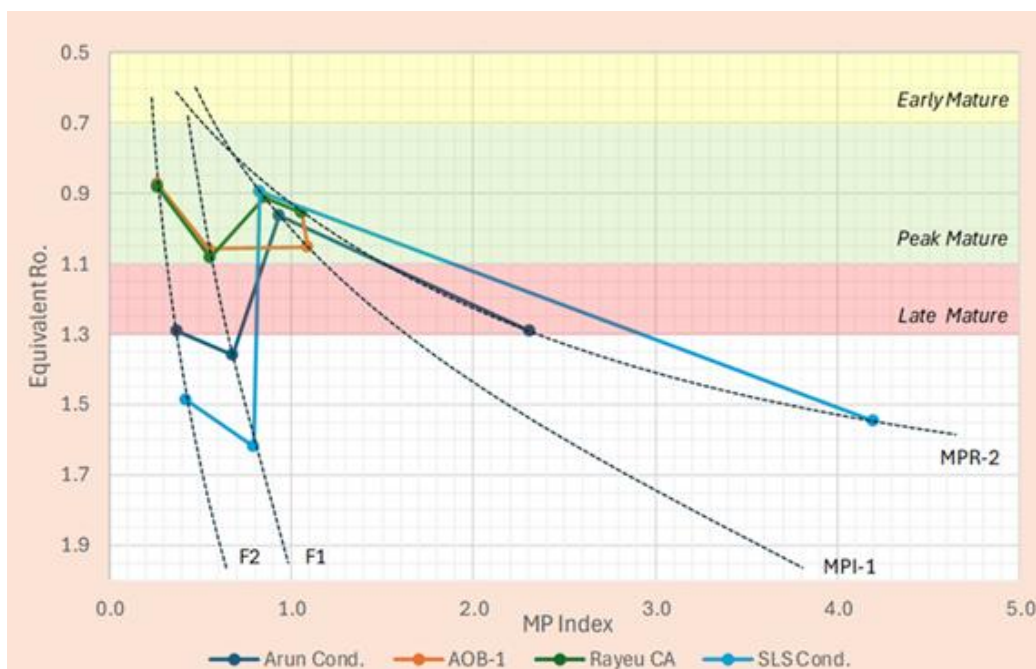


Figure 13.

Thermal maturity assessment based on the Methylphenanthrene index (MPI), indicating high condensate maturity with an equivalent vitrinite reflectance (Ro) ranging from 0.88 to 1.08%.

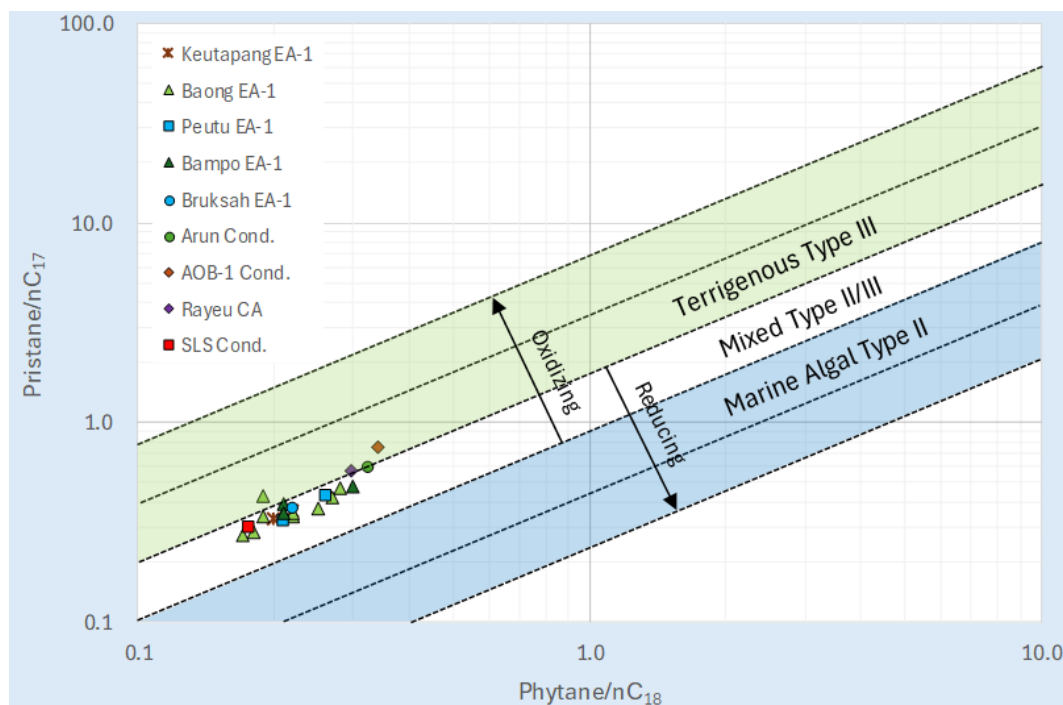


Figure 14.

Crossplot of Phytane/ $nC_{18}$  to Pristane/ $nC_{17}$  from condensate sample and rock extract of East Arun A-1 well, indicating deposition under oxic to suboxic conditions.

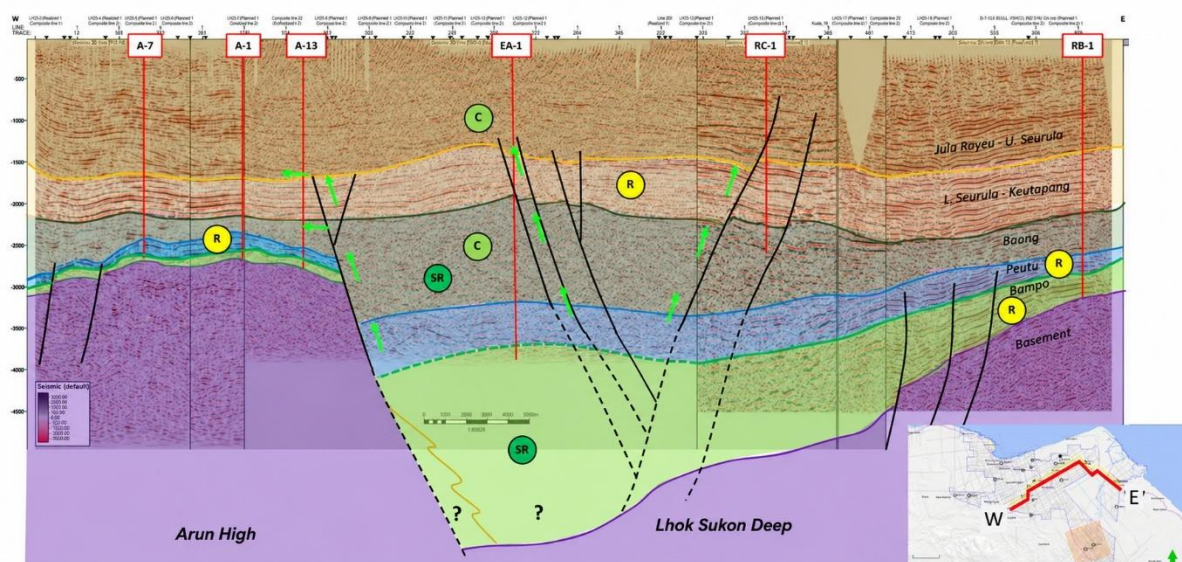


Figure 15.

Interpreted relatively east-west seismic section across Lhok Sukon Deep sub-basin. The deepest geochemical data were acquired from East Arun A-1, which only penetrated the top of the Bampo Formation.

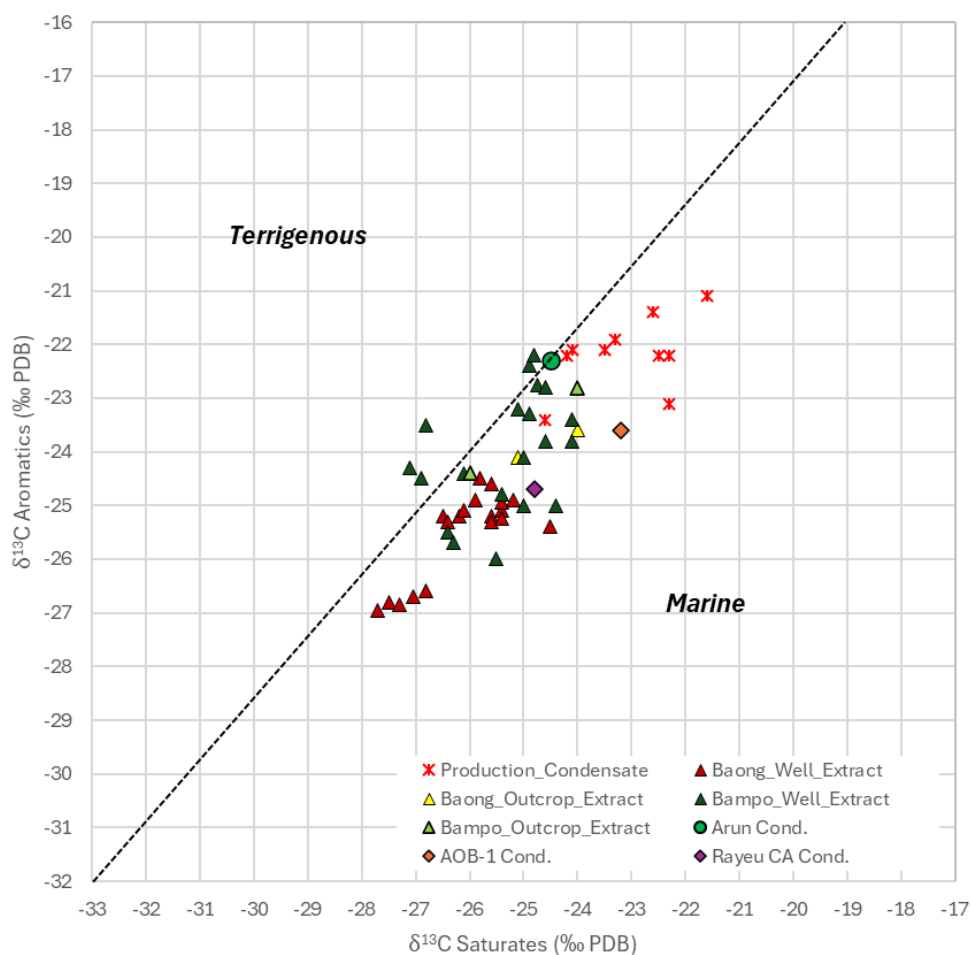


Figure 16.

Cross-plot of carbon isotope saturation against aromaticity.

The condensate and rock extract samples generally indicate derivation from marine to mixed marine-terrestrial organic matter. Condensate samples exhibit a closer affinity to the Bampo Formation rock extracts than to the Baong Formation rock extracts, which are separated by approximately 1-4‰.

#### 4. Conclusion

The North Sumatra Basin, particularly the Lhok Sukon Deep sub-basin, contains several potential source rock intervals; however, geochemical evaluation of source rock richness, kerogen quality, and thermal maturity suggests that the Baong, Peutu, Bampo, and Bruksah Formations constitute the most viable source rock candidates. Gas and condensate samples indicate generation from a highly mature petroleum system. The retrograde nature of the condensates further suggests derivation from Type II to Type II/III kerogen at advanced stages of thermal maturity. This confirms that an effective petroleum system was capable of charging younger reservoirs.

Geochemical parameters, including biomarker distributions and Pr/Ph ratios, indicate that the source organic matter was deposited predominantly in a marine setting under anoxic to suboxic conditions, with only a minor terrestrial contribution. The biomarker signatures also support a shale-dominated source facies.

Carbon isotope data demonstrate that condensates from the Rayeu CA and AOB-1 wells are genetically related to the Bampo Formation shale source rock. Condensates from the Arun Field exhibit similar isotopic and molecular signatures, suggesting a common source origin. This interpretation is further supported by the close similarity between the n-alkane distributions of the condensates and those of extracts from the Bampo Formation. Consequently, gas and condensate samples from the Lower Seurula sandstone reservoir are most consistent with a highly mature Late Oligocene Bampo-type marine shale source facies. However, hydrocarbon migration from other pre-Baong potential source rock intervals could not be entirely ruled out.

The recent gas and condensate discoveries within the Lower Seurula Formation sandstone reservoirs provide compelling evidence for an active petroleum system in the Lhok Sukon Deep Sub-basin, North Sumatra Basin. The results indicate that hydrocarbons generated from the Bampo Formation marine shale source rock have migrated into both the Peutu carbonate reservoirs and the stratigraphically younger Lower Seurula sandstone reservoirs. This migration is interpreted to have occurred through a combination of fault-related pathways and permeable carrier beds, enabling vertical and lateral hydrocarbon charge across multiple stratigraphic intervals.

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#### 6. Declarations

##### 6.1 Author contributions

W.A. Faridsyah: Conceptualization, data analysis, investigation, visualization, writing – original draft. V. Susanto: Supervision, technical review, validation, writing – review and editing.

##### 6.2 Funding statement

The research did not receive any funding from any external organization or party.

##### 6.3 Competing interest

The authors declare no known competing interests that could influence the research.

##### 6.4 The use of AI or AI-assisted technologies

The authors used Microsoft Copilot for language refinement, then reviewed and edited it to avoid any discrepancy with the geoscientific context.

## 7. Glossary of terms and symbols

| Terms & symbols       | Description                                                          | Unit |
|-----------------------|----------------------------------------------------------------------|------|
| TOC                   | Total organic carbon                                                 | -    |
| PVT                   | Pressure-volume-temperature                                          | -    |
| REP                   | Rock-eval pyrolysis                                                  | -    |
| MD                    | Measured depth                                                       | -    |
| GC-MS                 | Gas chromatography – mass spectrometry                               | -    |
| NSO                   | Nitrogen, sulfur, oxygen                                             | -    |
| SANA                  | Saturate, aromatic, nso, asphaltene                                  | -    |
| m/z                   | Mass-to-charge ratio                                                 | -    |
| °C                    | Celsius temperature unit                                             | -    |
| °F                    | Fahrenheit temperature unit                                          | -    |
| $\delta^{13}\text{C}$ | Stable carbon isotope ratio                                          | -    |
| ft                    | Feet, length unit                                                    | -    |
| Ro                    | Vitrinite reflectance                                                | -    |
| T <sub>max</sub>      | Maximum temperature at s <sub>2</sub> in rock-eval pyrolysis         | -    |
| wt. %                 | Weight percentage in rock                                            | -    |
| %Ro                   | Percentage of light reflected from the surface of polished vitrinite | -    |
| ‰                     | Permil or per thousand                                               | -    |

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